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The Director
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Through Proper Channel

Respected Sir,

Sub: Student Innovative Project 2024 (SIP ID 2425S0141) - Submission – reg.

--*-*-*

We are herewith attached one copy of Student Innovative Project SIP 2024 (SIP ID 2425S0141) titled as "Swap and Play: A Motion Activated Smart Bluetooth Speaker" for your kind perusal. The same copy has been submitted in ctdt.annauniv.edu. Kindly request you to consider and do the needful.

Thank you

Yours Sincerely,

.2024
ercoil

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A.Anto Jeba Lispen
R.K. Adharsh)

AS Abhi.
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Adharsh

Mentors:
(Dr.V.A.Nagarajan and Dr.J.Banumathi)

Encl.: One hard copy of project proposal

UNIVERSITY COLLEGE OF ENGINEERING
ANNA UNIVERSITY CONSTITUENT COLLEGE

Forwarded

DEAN
University College of Engineering
Anna University Constituent College

PROPOSAL REPORT

Swap and Play : A Motion Activated Smart Bluetooth Speaker

Submitted by

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BONAFIDE CERTIFICATE

Certified that this proposal report " **Swap and Play : A Motion Activated Smart Bluetooth Speaker** " is submitted by *Mr. A S. Abhilash*, *Mr. A. Anto Jeba Lispen*, *Mr. R.K. Adharsh*.

We certify to read and follow the guidelines prescribed by the University during and after implementation of the project.

SIGNATURE OF STUDENTS

1. Abhilash
Mr. A S. Abhilash

2. Lispen
Mr. A. Anto Jeba Lispen

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SIGNATURE OF MENTOR

1. ____________________
Dr. J. Banumathi

2. ____________________ 25/10/24
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RECOMMENDED BY HOD / DIRECTOR

25/10/24

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PROJECT DETAILS

OBJECTIVES

Objectives:

· To reduce the electronic waste by battery replaceable smart Bluetooth speaker rather than replace the entire gadget.

· To provide beneficial for residences, pool parties, and outdoor events and gatherings with a smart portable Bluetooth speaker

· To gain access through gestures for a more engaged experience while drive the car.

· To keep the product affordable for customers, strike a balance between cost considerations and the use of sustainable materials and technology.

INTRODUCTION

Bluetooth speakers are widely used for their portability and wireless convenience. However many existing devices are having fixed batteries, which limit their operational time or require the speaker to be plugged in for charging thus reducing portability. There is a need for a Bluetooth speaker that combines the convenience of a removable battery with the ability to charge the battery both inside and outside the speaker. Bluetooth device is very essential for car drivers while driving the car. Most of the yoga practitioners using this device to connect to the mobile or systems while doing their daily routine exercises. So it is a important device in every place. Hence it is proposed to a smart Bluetooth speaker with removable and rechargeable battery.

The present invention relates to portable audio devices, particularly to Bluetooth speakers with advanced battery management features, including are movable and rechargeable battery system.

LITERATURE SURVEY

Bluetooth speakers have gained immense popularity due to their portability and ease of use, enabled by wireless connectivity. One of the primary concerns with portable speakers is the power source, which typically comes from rechargeable batteries. In recent years, there has been an increasing interest in speakers with removable and rechargeable batteries to enhance convenience, longevity, and sustainability. This survey explores existing models of Bluetooth speakers with removable batteries and examines relevant research literature in the field.

Modular Bluetooth Speaker Design:

Many modern Bluetooth speakers are designed with modular components, including the battery, to improve product longevity and user convenience. This allows users to easily swap out or replace batteries, extending the overall lifespan of the device.

A study by Zhang et al. (2018) investigated the feasibility of modular designs in Bluetooth audio devices, specifically focusing on the ease of battery replacement. The research found that modular designs increase the product's reparability and reduce electronic waste, supporting sustainability efforts in consumer electronics.

Rechargeable Lithium-ion Batteries in Bluetooth Speakers:

Most Bluetooth speakers utilize lithium-ion batteries due to their high energy density, rechargeability, and long cycle life. Research has focused on optimizing the design of these batteries to support high power output while maintaining safety and longevity.

In this paper by Kumar et al. (2019), explored the efficiency and performance of lithium-ion batteries used in portable Bluetooth speakers. The study highlighted improvements in battery life, charging time, and thermal management, which are critical for ensuring user safety and extending battery longevity.

Battery Management Systems (BMS)

A critical aspect of Bluetooth speakers with removable and rechargeable batteries is the battery management system (BMS), which controls the charging, discharging, and health of the battery. BMS ensures that the battery operates within safe parameters, prolonging its life and enhancing user safety.

A paper by Lee et al. (2020) discussed the role of advanced BMS in portable Bluetooth devices with removable batteries. The research analyzed how BMS technologies monitor battery health and prevent overcharging or overheating. The authors noted that advancements in BMS have significantly contributed to the safety and durability of Bluetooth speakers, especially those with user-replaceable batteries.

Consumer Preference for Removable Batteries:

In a study examining consumer preferences, researchers have noted that users often prefer portable electronics with removable batteries. This preference stems from concerns about battery degradation over time, which can render a device unusable if the battery is not replaceable.

A survey conducted by O'Neil et al. (2021) explored consumer attitudes toward portable electronics, including Bluetooth speakers. The results showed that 65% of consumers favored devices with removable batteries, citing the ability to replace the battery as a major factor in extending device usability and reducing waste.

Sustainability and Environmental Impact:

The environmental impact of electronic waste (e-waste) has led to increased interest in designing products with removable components, including batteries, to improve their repairability and recyclability. This approach aligns with circular economy principles, which emphasize reducing waste and maximizing resource use.

A study by Venkatesh et al. (2022) examined the environmental benefits of portable Bluetooth speakers with removable batteries. The research found that products with easily replaceable batteries contribute to reducing e-waste and promote longer product life cycles, aligning with sustainable design practices. The study advocated for regulatory measures encouraging manufacturers to adopt removable battery designs in portable electronics.

The literature suggests that Bluetooth speakers with removable and rechargeable batteries offer several advantages, including enhanced convenience, sustainability, and product longevity. Research highlights improvements in battery technology, modular design, and battery management systems, all of which contribute to the growing popularity of such devices. Further developments in battery efficiency and sustainability practices are expected to drive the future of Bluetooth speaker design.

Inferences:

The built-in battery of the existing Bluetooth speaker is non-replaceable. This reduces the Bluetooth speaker's mobility.

We have improved this by adding a detachable battery to the speakers. It is possible to remove this battery at any time.

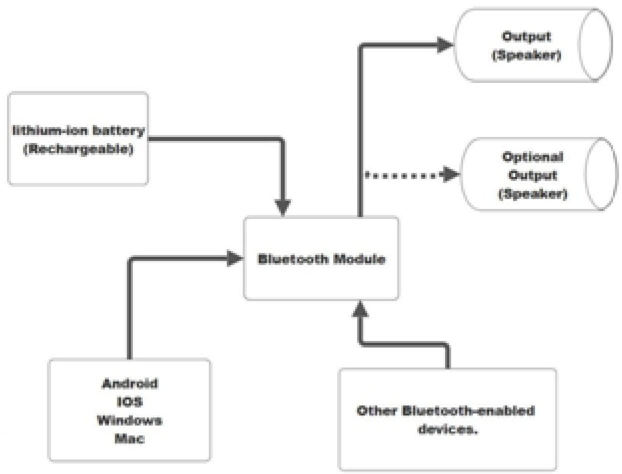
Instead of throwing away the speaker altogether, customers can change the battery as it becomes inefficient. This promotes sustainability by extending the device's lifespan and lowering electronic waste. While comparing with the existing speaker, Swap and Play speakers are more eco-friendly.

Our speaker drastically reduces the price of the existing model, which features a removable battery.

The existing speaker lacks gesture control and can only be operated physically or with a remote. However, we have added sensors to enable gestures, which improves the speakers' logical interaction.

PROPOSED WORK WITH METHODOLOGY

Block Diagram



The diagram above is the block diagram of the project. This represents a simple layout for the input and output lines from the power supply (lithium-ion battery), Bluetooth module, speakers, controlled over Android, iOS, Windows, Mac, and other Bluetooth-enabled devices.

Power supply:

The power supply 3.7v was given to the Bluetooth Module. The module requires a constant 5v D.C supply for the smooth operation. The speakers get their range of supply from the module as required for output.

Bluetooth Module:

This is a perfect Bluetooth board module as to set a simple audio output system comes on a single board. This module supports Bluetooth, USB, mp3, aux. cable etc.This mini-module is very easy to install. And also it has a dial switch for switching applications like volume, stop, etc.

Features:

a. It consists of inbuilt memory card slot and decoder

b. It works well with 5V DC current

c. It can simply connect a speaker to it without any problem

d. It supports Bluetooth, USB, aux cable, memory chip decoder,

e. It has a dial for switching applications

f. It is very cheap and easy to install

Speakers:

An 8Ω 3W speaker is a common type of speaker used in low-power audio applications like small radios, Bluetooth speakers, or DIY projects.

Characteristics:

1.Impedance: 8Ω (Ohms);

· Impedance refers to the resistance the speaker provides to the current from the amplifier.

· 8Ω is a standard impedance for many audio speakers, meaning the speaker will provide 8 ohms of resistance.

2.Power Rating: 3W (Watts);

· The power rating indicates the maximum continuous power the speaker can handle without getting damaged.

· A 3W rating means the speaker can handle up to 3 watts of power from the amplifier.

3.Speaker Performance;

· **Volume:** A 3W speaker is not extremely loud but can produce enough volume for small spaces, such as desktop setups or portable speakers.

· **Efficiency:** Speakers with a higher efficiency rating (measured in dB) will produce more sound for the same amount of power compared to less efficient speakers.

Hardware Components:

a. 18650 Lithium-ion Battery (3.7v ~ 5v,2000mAh)

b. Wireless HI-FI Bluetooth Module Board

c. Motion Sensor

d. 8ohm 3watt Speaker

e. Soldering Lead

f. PVC (Polyvinyl chloride)

g. Soldering Thread

IMPLEMENTATION

Working Principle:

At first the input is taken from the rechargeable battery and they were given to the Bluetooth board. The board accepts only rated voltage of 5v if the voltage exceeds beyond the rated level, the Bluetooth's signal plate got heated and burn out.

Hence the 18650 Lithium-ion Battery (3.7v ~ 5v, 2000mAh) holds this level of voltage, which helps to run. In the next phase, the Bluetooth passes the data signals to the speakers, which were connected on the board port. Here the speaker contains ratings of 8 ohms and 3 watts.

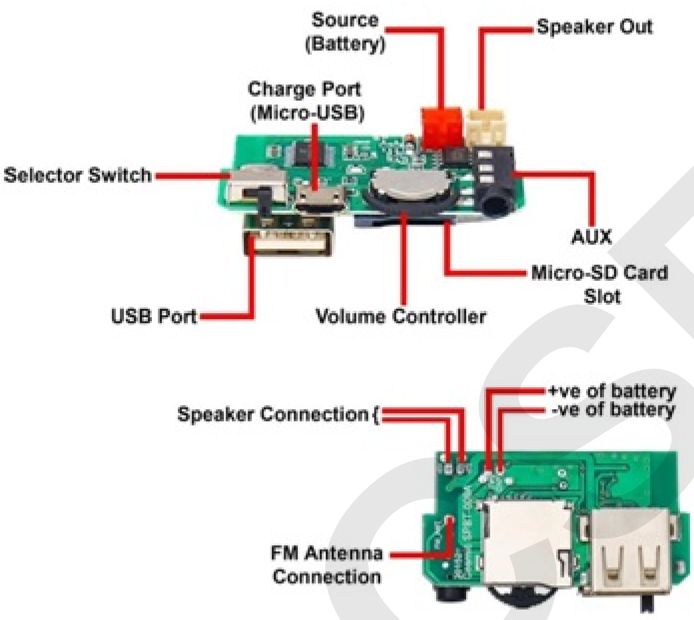
Then the Bluetooth was enabled, and the data were transmitted. Here the Android plays the role of transmitter and the Bluetooth plays the receiver role.

At this state, a motion sensor was given to get easy access to run the module without any external remote. The sensor accepts the analog inputs, passes a digital signal, and runs based on the logical operation.

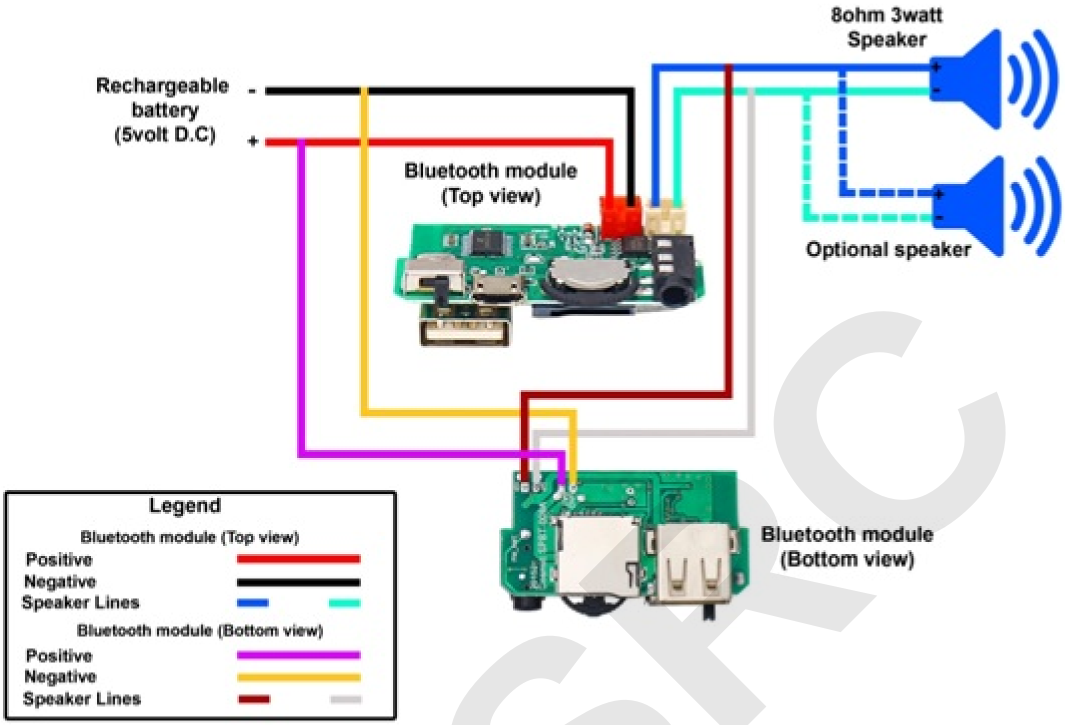
When the sensor accepts the analog signal (hand gesture), it acts based on the movement that was fed. Based on this, the music will get play, pause, repeat, shuffle, on /off. The board and circuit diagrams

are given below.

Board Details:



Circuit Diagram:



WORK PLAN

a. Literature survey : November 1 to November 15

b. Hardware and Software purchase: November 16 to November 25

c. Design and Analysis: November 26 to December 15

d. Implementation : December 16 to January 15

e. Result: January 16 to February 15

f. Report Preparation: February 16 to February 28

EXPECTED OUTCOME / RESULTS

T It reduces the electronic waste by battery replaceable smart Bluetooth speaker rather than replace the entire gadget.

· beneficial for residences, pool parties, and outdoor events and gatherings with a smart portable Bluetooth speaker

· gain access through gestures for a more engaged experience while drive the car.

· the product is affordable for customers, strike a balance between cost considerations and the use of sustainable materials and technology.

APPLICATIONS

ü

Fitness and Wellness

ü Children's Play Areas

ü Educational environment

ü Home Automation

ü Outdoor events

a Car drivers

CONCLUSION

In summary, the Bluetooth speaker with a motion sensor and detachable battery presents an alluring combination of innovation and convenience. By making it simple for users to replace drained batteries and removing the need to wait for recharging, the detachable battery guarantees continuous use. In the meantime, the built-in motion sensor improves user engagement by triggering playback automatically as users go closer, allowing for hands-free use.

Because it allows battery replacement rather than whole device disposal, this design is not only user-friendly but also environmentally responsible. All things considered, this Bluetooth speaker is a flexible audio option that satisfies the needs of contemporary lives, making it perfect for a range of settings, including homes and outdoor parties. This initiative establishes a new benchmark in the portable audio industry by putting functionality, sustainability, and user involvement first.

REFERENCES

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FINANCIAL ASSISTANCE	
HEAD	**AMOUNT(Rs.)**
Material / Fabrication / Component	16000
Travel	4000
Contingency	2000
Consumables	0
Others	2950
TOTAL	**24950**

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